

Semantic Segmentation

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24.11.08

Introduction

- ☐ Semantic Segmentation

Architecture

- ☐ FCN
- ☐ UNet

Experiment

- ☐ Evaluation Metrics
- ☐ Result

Conclusion

- ☐ Future Work

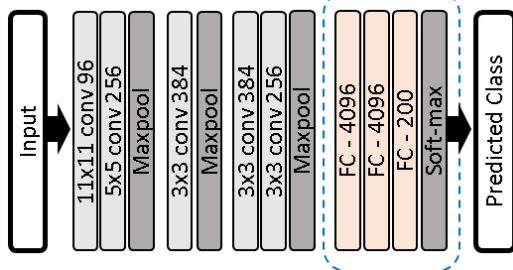
■ FCN^[1]

- Main features of the network

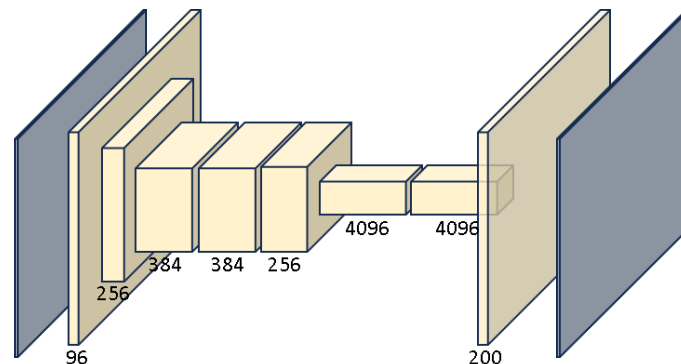
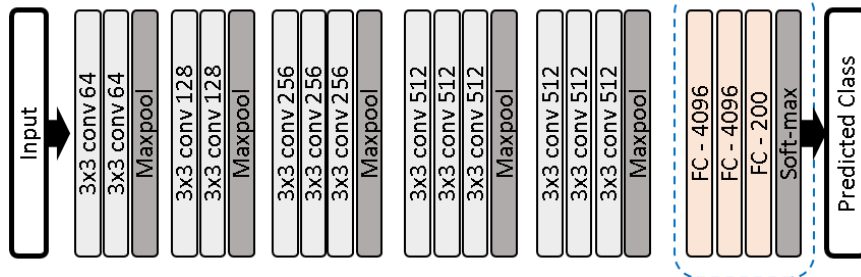
1. Adopts a classification network
2. Changes FC layers to convolution layers
3. Combines information from a deep layer with that from a shallow layer

<Classification Networks>

• AlexNet



• VGG net



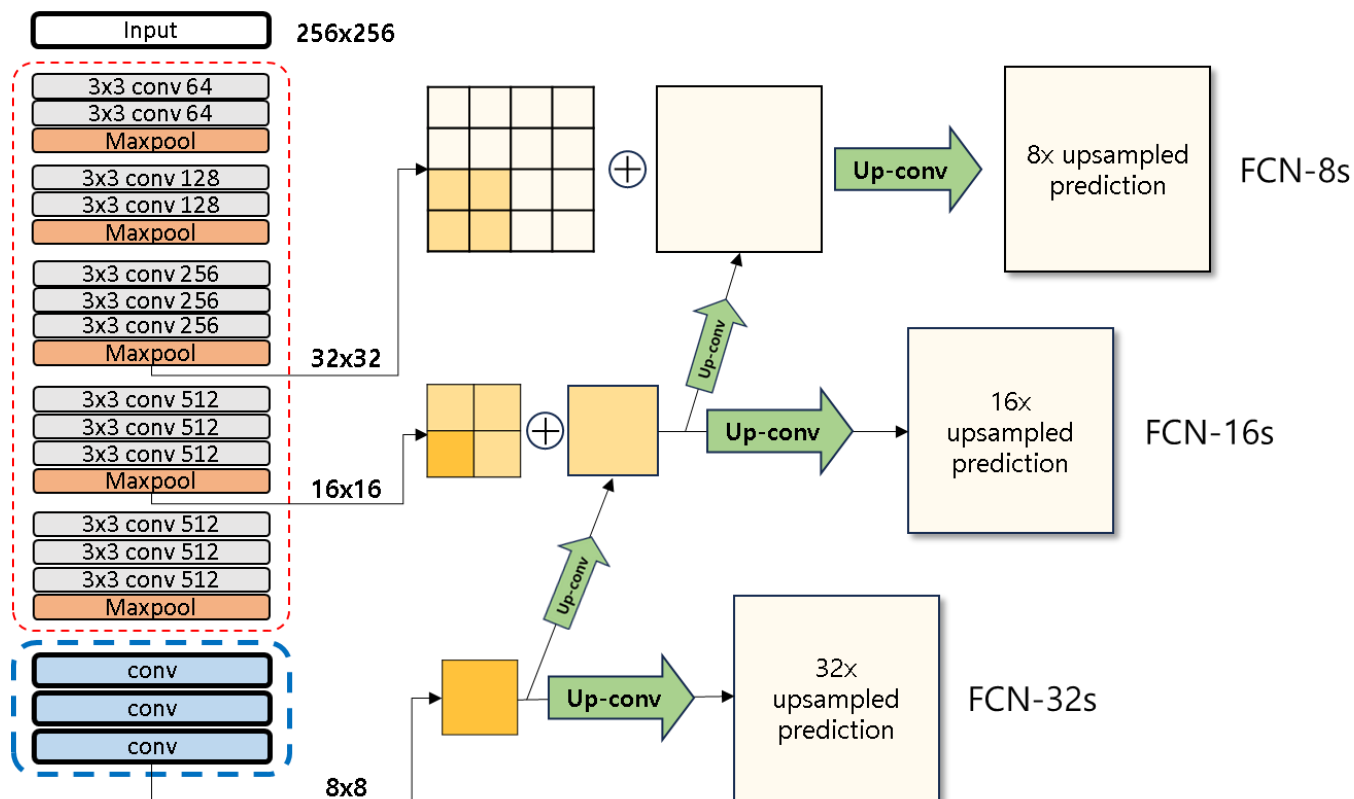
<Feature Map Shapes After Convolutionalization>

[1] Long, J. et al. "Fully Convolutional Networks for Semantic Segmentation." *Computer Vision and Pattern Recognition*. 2015.

FCN

- Main features of the network

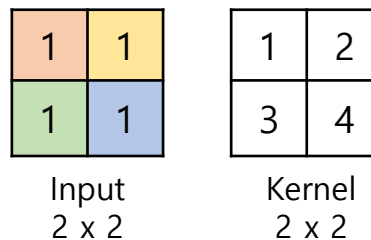
1. Adopts a classification network
2. Changes FC layers to convolution layers
- ③ Combines information from a deep layer with that from a shallow layer



▪ FCN

- Up-sampling

- Transposed Convolution: Learnable parameters → enables more refined feature extraction



a. stride:2, padding=0

1	2	1	2
3	4	3	4
1	2	1	2
3	4	3	4

Output
4 x 4

* Multiply each kernel element with a single input pixel

b. stride:1, padding=0

1	3	2
4	10	6
3	7	4

Output
3 x 3

* Slide kernel by stride

c. stride:2, padding=1

1	2	1	2
3	4	3	4
1	2	1	2
3	4	3	4

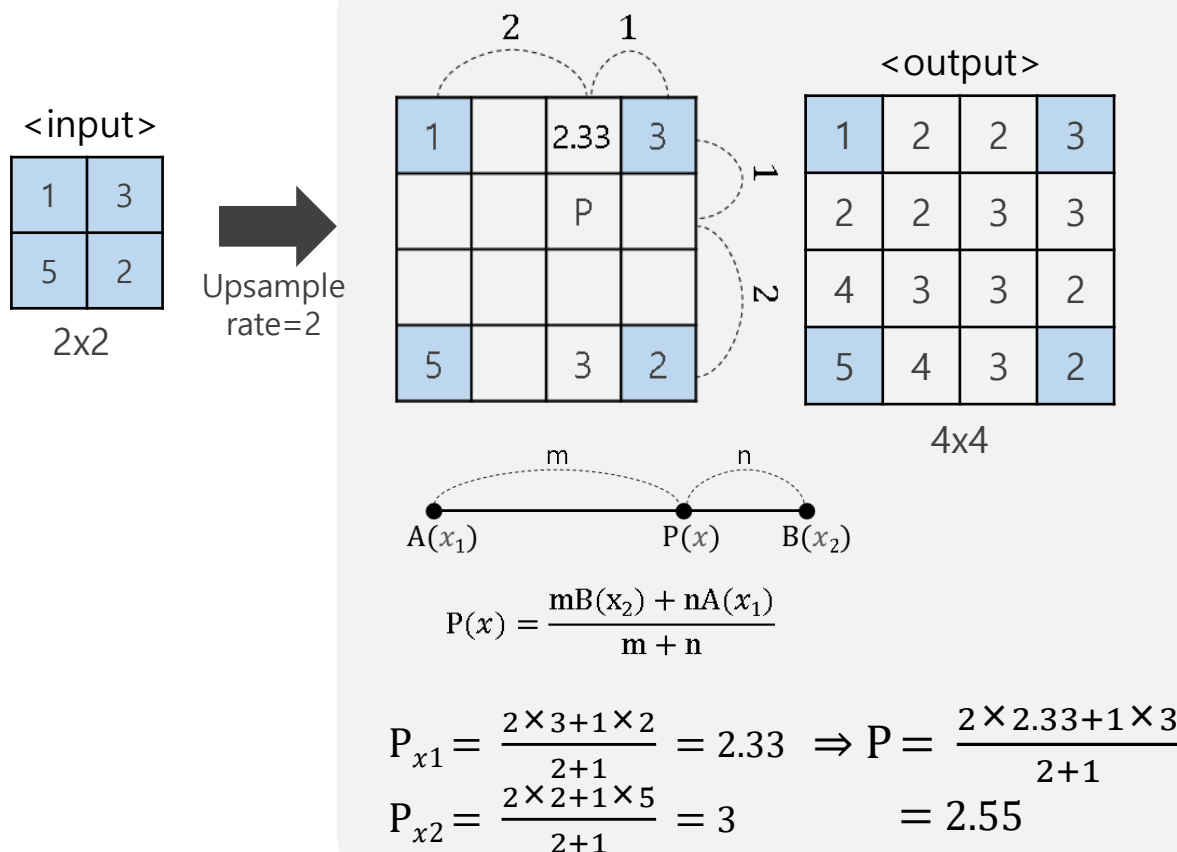
Output
2 x 2

* Remove elements by padding value

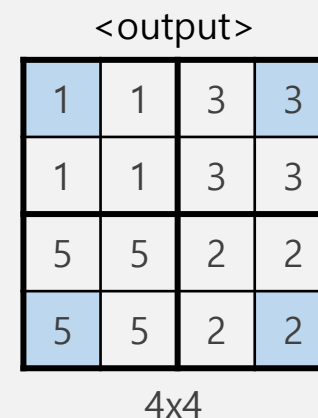
FCN

- Up-sampling

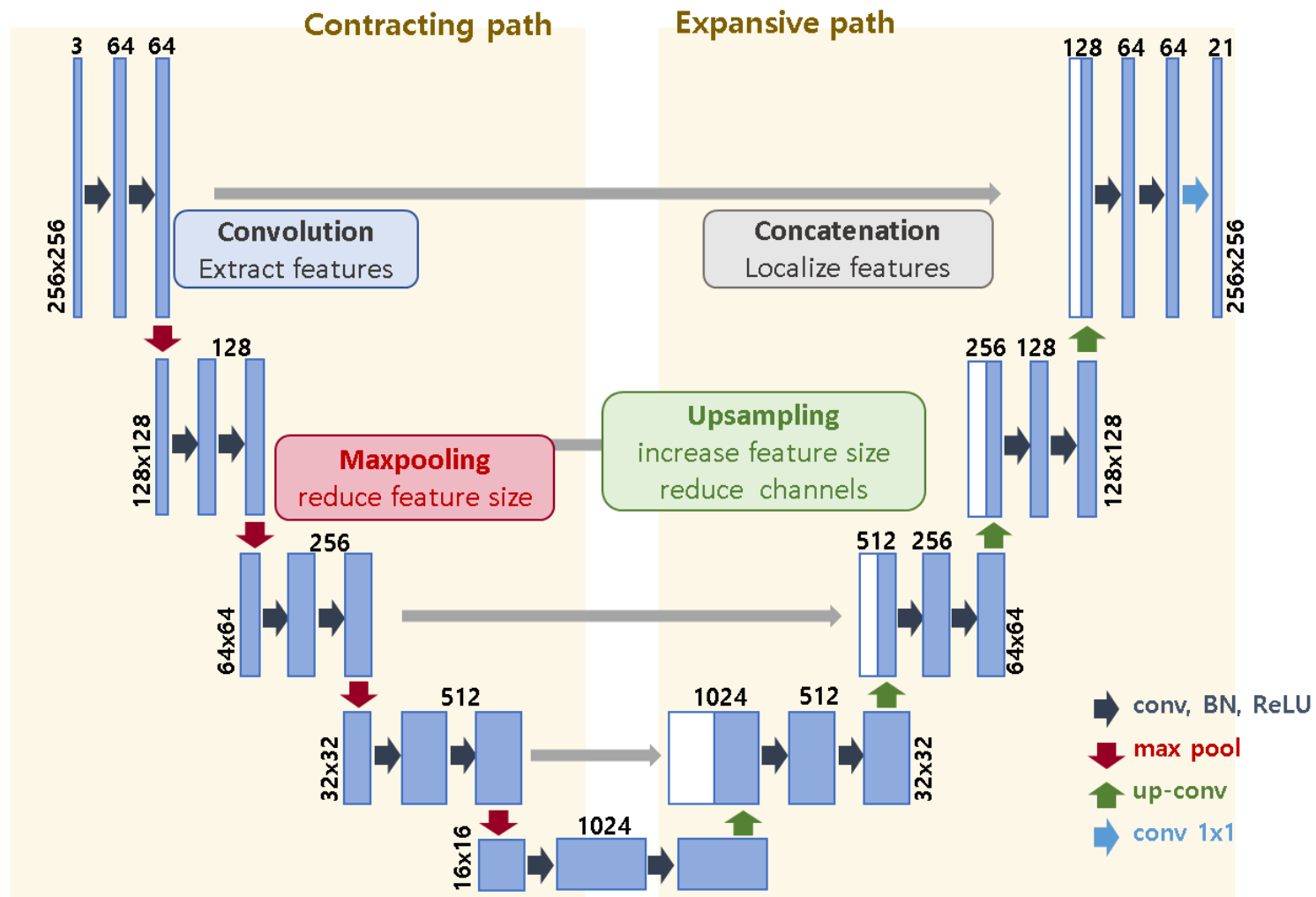
Bilinear Interpolation



Nearest-Neighbor Interpolation



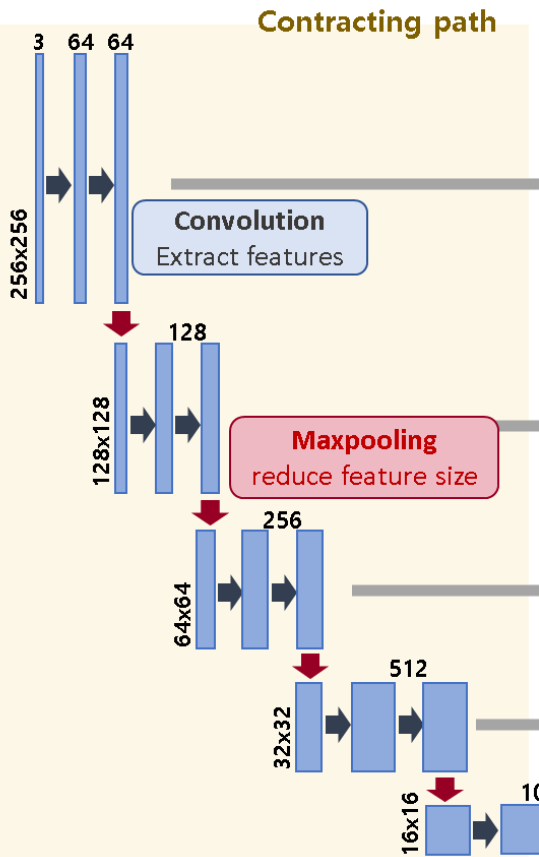
■ UNet^[2]



[2] Ronneberger, O. et al. "U-Net: Convolutional Networks for Biomedical Image Segmentation." *Medical Image Computing and Computer-Assisted Intervention*. 2015.

UNet

- Contracting path



Convolution

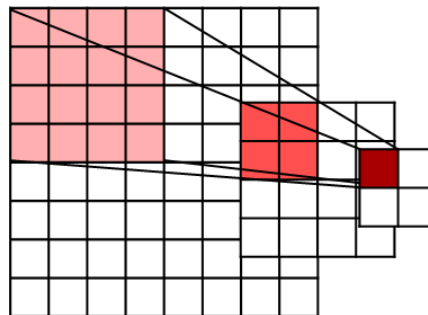
Extract features

- Extract various features increasing channels

Maxpooling

Reduce feature map size

- Reduce feature map size
➡ Capture context information



Lines, Corners, Edges

Meaningful Faces

Feature map size ↓



Receptive field size ↑



Include **context** information

Lower layer : **basic** features

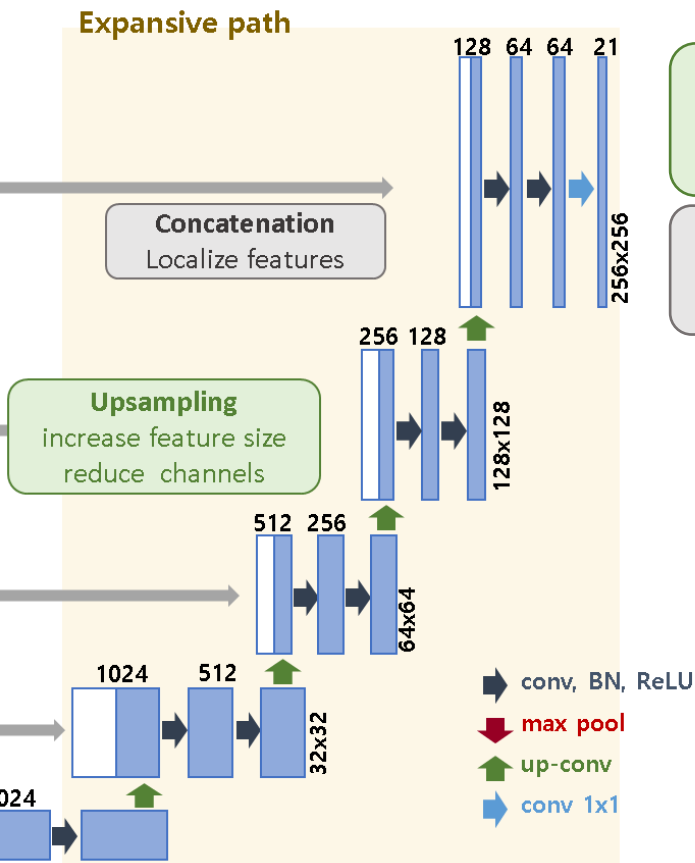
e.g. edges, corners, contours ...

Higher layer : **complex** patterns

e.g. textures, object parts ...

UNet

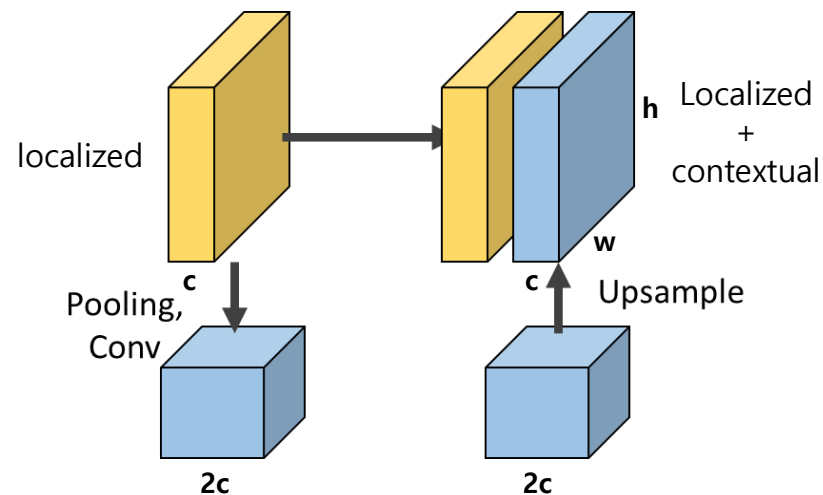
- Expansive path



Upsampling
twice height, width
half channel

Concatenation
Localize features

- Restore resolution
- Decrease the channels
- Connect upsampled feature map with one of the same size from contracting path
➔ Includes the missing localized features

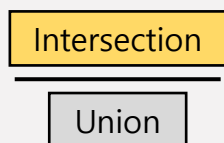
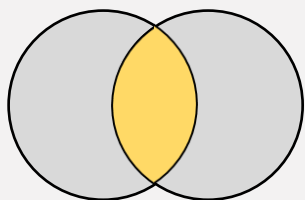


■ Evaluation Metrics

- mIoU : mean Intersection over Union
- Pixel Accuracy

mIoU

IoU: Intersection over Union



0	1	0	0	2
1	1	1	0	2
1	1	1	0	0
0	1	0	0	0
0	1	0	0	0

Ground truth

0	0	0	2	2
1	1	1	0	2
0	1	2	0	0
0	1	1	0	0
0	1	0	0	0

Prediction

		Ground truth		
		0	1	2
Prediction	0	12	2	1
	1	1	6	0
	2	0	1	2

per-class IoU

class 0: $\frac{12}{16}$, class 1: $\frac{6}{10}$, class 2: $\frac{2}{4}$

$$\text{mIoU} = \left(\frac{12}{16} + \frac{6}{10} + \frac{2}{4} \right) / 3 = \mathbf{0.62}$$

Pixel Accuracy

		Ground truth		
		0	1	2
Prediction	0	12	2	1
	1	1	6	0
	2	0	1	2

Pixel accuracy: $\frac{20}{25} = \mathbf{0.8}$

▪ Result

Set up

	FCN_8s	Unet
Dataset	PASCAL VOC2012	PASCAL VOC2012
iteration	100,000	80,000
Batchsize	32	32
Learning rate	0.001	0.0001
Learning rate scheduling	Every 50,000 iterations, reduce by half	Every 35,000 iterations, reduce by half
Optimizer	Adam	Adam
Criterion	Cross Entropy Loss	Cross Entropy Loss
Augmentation	Horizontal flip, 45° rotation	Horizontal flip, 15° rotation, Reduce image saturation to 70%
Hardware	NVIDIA GeForce RTX 4090	

Result

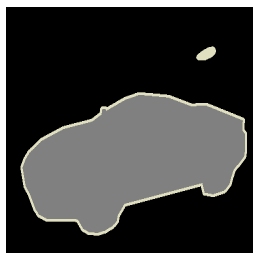
FCN_8s

	Pixel Accuracy	mIoU
FCN_8s	0.84	0.38
UNet	0.81	0.26

Input image



Ground Truth



FCN_8s



0.98	0.96
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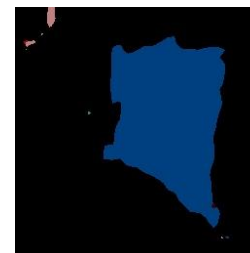
Input image



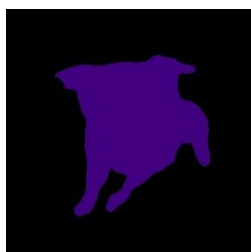
Ground Truth



FCN_8s



0.17	0.09
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0.98	0.95
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0.04	0.01
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Result

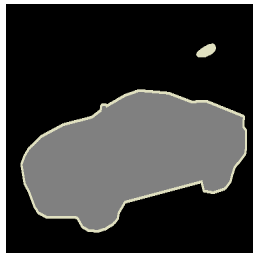
UNet

	Pixel Accuracy	mIoU
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Input image



Ground Truth



UNet



0.99	0.97
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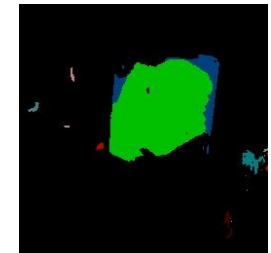
Input image



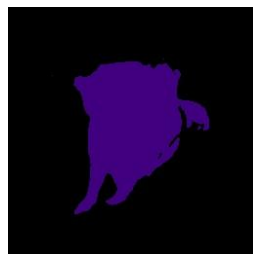
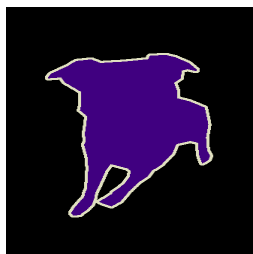
Ground Truth



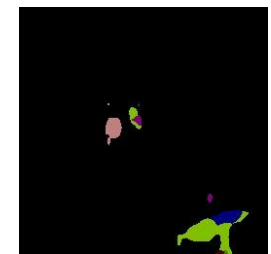
UNet



0.19	0.32
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0.96	0.85
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0.03	0.33
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Thank you